

Arpan Pathak

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📄 SUMMARY

Senior Software Engineer, **8 years** building developer tools, observability platforms, and performance-critical distributed systems at **Amazon, Microsoft, Oracle**. Full-stack **Rust, C++, Kotlin, Go, Python, JavaScript; Kubernetes, eBPF/Cilium**, GPU/**CUDA**, cloud **performance optimization**.

🏢 EXPERIENCE

Software Developer II, Microsoft, Redmond, WA

Jun 2025 - Aug 2026

- **Cluster Observability, K8s & eBPF: Prometheus + Azure Monitor**, blue-green deployments, Kubebuilder operators, eBPF/Cilium L7 filtering across **AKS** clusters in France and Germany.
- **C++ → Rust Migration**: Rebuilt Microsoft Information Protection from **legacy C++/C#** in **Rust**, moving memory safety to compile time; ~ **40%** lower P99, **2x throughput**, **3x lower memory/request**.
↳ TechStack: **Rust, C++, eBPF, Cilium, Kubernetes, Linux**

Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA

Oct 2024 - Mar 2025

- **Developer-Facing SDK (Go)**: Built the **Terraform provider** for Oracle Autonomous Database, replacing legacy control-plane tooling; **\$1.2M/month projected savings**.
- **Database Engine Internals**: Optimized **lock-free data structures** in a proprietary DB engine, achieving + **45,000 ops/s** in high-concurrency read-write paths.
↳ TechStack: **Java, Go, Terraform, OCI**

Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India

Mar 2021 - Oct 2024

- **Real-Time ML Ranking (Python/PyTorch)**: **XGBoost + deep-learning ensemble** at **100ms** across FreeVee, Twitch, Prime Video; **5% CTR lift**, **\$5.4M/month revenue growth**.
- **BERT NLP Service**: **5,000 QPS** at peak; cut support-ticket triage SLA from **7 days** to **2 hours**.
- **Full-Stack Developer Platforms**: **React.js microfrontend** migration (deploy **5 days** → **30 min**); event-bus **SDK** (Kotlin, < **50ms**) used by publisher teams.
- **Data Platform**: **1-petabyte** warehouse/lake with **Spark (AWS Glue)** pipelines for financial reporting and BI.
↳ TechStack: **Kotlin, Java, Python, PyTorch, Spark, Kafka, DynamoDB, Redshift, AWS**

SDE, Razorpay, Bengaluru, India

Aug 2020 - Feb 2021

- Go UPI Payment Gateway: **1M+** txns/hr, **99.99%** uptime, **150ms** P99 at 50K TPM.

Software Engineer, Mindfire, Bhubaneswar, India

Aug 2018 - Feb 2020

- Java/Spring full-stack tools, AR product-search microservice (**10K+** daily queries); latency **5 min** → **1 min**.

📄 PERSONAL PROJECTS

🚗 Driving-CivicSense: Edge AI Vision System (Rust)

- Real-time on-device perception pipeline in **Rust** on the **NVIDIA Jetson Orin Nano Super** (67 INT8 TOPS): **YOLOv8/v11 INT8 ONNX** (~ **12 ms**) + Deep SORT/Kalman + proven kinematic engine; **100%** on-device, privacy-first.
- Research paper: **Deterministic Intersection Blockage Prediction** | arpanpathak.github.io/driving-civicsense-vision-model/ | Code: github.com/arpanpathak/driving-civicsense-vision-model
- Books: **Seeing Machines** | arpanpathak.github.io/seeing-machines-book · **CUDA Kernels** (CUDA-Oxide / Rust-to-CUDA) | arpanpathak.github.io/gpu-parallel-book

🔧 SKILLS

↳ **Languages**: Rust, C++, Kotlin, Go, C, Java, Python, TypeScript, JavaScript, Shell Script

🔧 **Systems Programming & Performance**: Lock-Free Data Structures, Concurrency, Memory Safety, Memory Profiling, Cache Optimization, Performance Engineering, Low-Latency Systems, OS Internals, Computer Architecture

🌐 **Networking & Protocols**: TCP/IP, UDP, TLS, HTTP/2/3, QUIC, RDMA, InfiniBand, DPDK, eBPF, XDP, Cilium, VPC, NAT, VPN, Load Balancing

📡 **Distributed Systems & Streaming**: Consensus, Replication, Sharding, Apache Kafka, Apache Flink, Apache Spark, Spark Streaming, AWS Kinesis, Change Data Capture (CDC), Microservices

☁ **Cloud & Infra**: AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS), Google Cloud, Kubernetes (EKS, AKS), Docker, Terraform, Kubebuilder, Istio

🔗 **ML Inference & AI**: Deep Learning, Computer Vision, PyTorch, TensorFlow, XGBoost, Transformer Models (BERT), LLMs, ONNX Runtime, YOLOv8/v11, CNN, NLP, Edge Inference; CUDA, CUDA-Oxide (Rust-to-CUDA), SIMT

📊 **Observability**: Prometheus, Grafana, Azure Monitor, AWS CloudWatch, Trace & Sampling Data, Performance Profiling

🗄 **Data & Storage**: DynamoDB, PostgreSQL, Redis, MongoDB, MySQL, Amazon Redshift, Amazon S3

🔗 **APIs & Serialization**: gRPC, REST API Design, Protobuf, React.js

🔧 **Core CS**: Data Structures & Algorithms, System Design, Operating Systems, Linux

🎓 EDUCATION

B.Tech, Computer Science & Engineering, RCC Institute of Information Technology, Kolkata | 2018